

AAYAN VIKRAM SINGH

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EDUCATION

PES University (EC Campus), Bengaluru, Karnataka Aug 2023 – May 2027
B.Tech, Computer Science and Engineering | CGPA: 9.34 / 10.00 (6 semesters completed)
• Prof. MRD Merit Scholarship — awarded every semester to date (6 of 6) for placing in the top 5% of the cohort.

Sainik School, Bijapur, Karnataka Jun 2015 – Jun 2022
Class XII: 94.4% | Class X: 91.2% | NCC 'A' & 'B' Certificates | NDA II 2022 — All India Rank 536

TECHNICAL SKILLS

Languages: Python, Java, C#, C, TypeScript / JavaScript, SQL
ML & AI: PyTorch, PyTorch Geometric, GNNs (TGN, HGT, GraphSAGE), RAG, FAISS, scikit-learn
Web & Services: .NET, gRPC, React, Node.js, Express, REST APIs, Supabase
Distributed & Data: Docker, RAFT, HDFS, PostgreSQL, MongoDB, MySQL, OSMnx / NetworkX
Tools: Git, GitHub Actions, Roslyn, Google Colab

EXPERIENCE & RESEARCH

AVEVA, Hyderabad Jun 2026 – Jul 2026
Software Development Engineer Intern
• Migrated aaBootstrapProxy — the gRPC gateway to the legacy COM AppServer subsystem — from 32-bit to 64-bit: audited 364 native dependencies and fixed 3 systemic COM/interop defects (marshalling, dynamic registration, struct layout).
• Validated on live System Platform environments — 45 tests passing with zero regressions — establishing the blueprint for migrating the remaining AppServer proxy services.
• Built Dev Insight Hub, a C#/Roslyn desktop tool pairing a .NET call-graph explorer with a multi-language dependency visualiser, adopted by the team for onboarding and change-impact analysis.

PESU Research Foundation, Bengaluru Jun 2025 – Present
Research Intern | Advisor: Dr. Richa Sharma
• Second author on “Modeling Online Radicalization Pathways using Temporal-Heterogeneous Graph Learning” — a Temporal Graph Network plus Heterogeneous Graph Transformer over 101,252 Reddit and 4chan posts.
• Trained multi-task heads for virality, hate-intensity, and early detection: AUC 0.85 vs a 0.87 DistilBERT baseline, with leakage-safe early detection at AUC 0.80 from a single interaction.

PES University — Chemistry Research Lab, Bengaluru Mar 2024 – Feb 2025
Research Assistant | Advisor: Dr. Revanasiddappa M | [Publication](#)
• Co-authored a published physicochemical assessment of groundwater quality across the BTM Layout industrial belt, analysing COD, TDS, pH, and conductivity across collected samples.

TEACHING EXPERIENCE

PES University, Bengaluru Aug 2026 – Present
Teaching Assistant — Machine Learning | Supporting labs, assignments, and doubt-resolution for the 5th-semester cohort

PES University, Bengaluru Jan 2026 – May 2026
Teaching Assistant — Computer Networks | Mentored the 4th-semester cohort through labs, assignments, and evaluations

PROJECTS

Agentic AI & GNNs for Sustainable Logistics | *PyTorch Geometric, OSMnx, LangGraph — Capstone* Aug 2025 – Present
Decentralised logistics platform where fleets and warehouses act as agents rerouting around live disruptions on GNN-predicted road costs.
• Built a Bengaluru road-network graph (7,095 nodes / 15,083 edges) from OpenStreetMap, plus 7,616 hourly training snapshots.
• Trained an edge-aware TransformerConv GNN predicting travel time, carbon emissions, and delay: R^2 0.77 / 0.79 against a GraphSAGE baseline's 0.59 / 0.49, with 35% lower validation loss.

SR-RAG — Self-Refuting Retrieval-Augmented Generation | *Python, FAISS, Groq, React* | [GitHub](#) Mar 2026 – Apr 2026
• Designed a multi-agent QA system routing low-confidence atomic claims to an adversarial Refuter constrained to retrieved evidence.
• Cut redundant LLM calls via dual-signal screening (model confidence + FAISS cosine similarity), logging every routing decision for claim-level auditability.

LeetTrack — LeetCode Analytics Platform | *React 19, TypeScript, Supabase / PostgreSQL, Python* | [GitHub](#) Sep 2025 – Nov 2025
• Built a peer-leaderboard platform on a 5-table PostgreSQL schema, fed by a GitHub Actions pipeline polling LeetCode's GraphQL API.

Mini-RAFT Collaborative Canvas | *TypeScript, Node.js, WebSocket, Docker* | [GitHub](#) Feb 2026 – Apr 2026
• Implemented RAFT leader election and log replication across three replicas so concurrent strokes commit only after majority acknowledgement, with automatic failover and catch-up sync.

Mini HDFS Simulation | *Python* | [GitHub](#) Oct 2025 – Nov 2025
• Recreated HDFS master-slave architecture: a NameNode owning namespace metadata and the file tree, with DataNodes handling block storage and replication behind a client interface.

BEYOND WORK

Half-marathon (first 21 km) · Trekking · Travelling · Painting · Reading (Apostrophe Literary Club) · Quizzing (QQC) · Theatre (Mangobites)